Employee Attrition Analysis Project

Exploratory Data Analysis & Business Insights Report

IBM HR Analytics Employee Attrition Dataset | 1,470 employees

1. Executive Summary

This report analyzes 1,470 employee records to understand the drivers of voluntary attrition and recommend actions to improve retention. Overall attrition stands at 16.1% (237 of 1,470 employees). The analysis shows attrition is heavily concentrated among a few identifiable groups — employees who work overtime, are early in their tenure, are single, or hold entry-level Sales/Lab roles — rather than being evenly spread across the workforce. This makes attrition a targetable, solvable problem rather than a broad cultural one.

KPI	Value
Total Employees	1,470
Employees who left (Attrition = Yes)	237
Overall Attrition Rate	16.12%
Overall Retention Rate	83.88%
Avg. Age — Left vs Stayed	33.6 yrs vs 37.6 yrs
Avg. Monthly Income — Left vs Stayed	₹4,787 vs ₹6,833 (units as in source data)
Avg. Tenure (Years at Company) — Left vs Stayed	5.1 yrs vs 7.4 yrs
Attrition Rate — Employees who work OverTime	30.5% (vs 10.4% for those who don’t)
Attrition Rate — New hires (<2 yrs tenure)	34.9%
Highest-risk segment	Single + OverTime employees → 49.6% attrition

2. Numerical vs Categorical Columns

For analysis purposes, the 32 core columns were classified as follows:

Type	Columns
Numerical (continuous)	Age, DailyRate, DistanceFromHome, HourlyRate, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, TotalWorkingYears, TrainingTimesLastYear, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager
Ordinal (rating scales, numeric-coded)	Education, EnvironmentSatisfaction, JobInvolvement, JobLevel, JobSatisfaction, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, WorkLifeBalance
Categorical (nominal)	Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime
Identifier	EmployeeNumber

3. Overall Attrition Overview

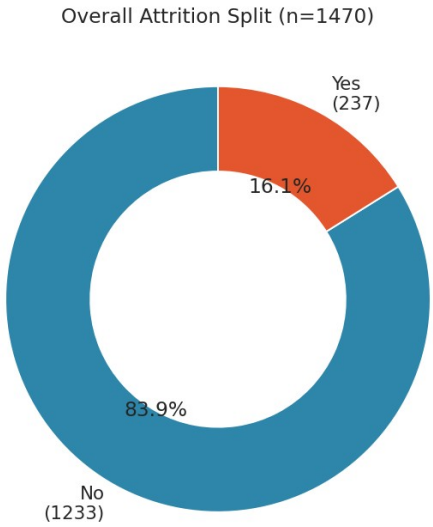


Figure 1: 237 of 1,470 employees (16.1%) left the company.

An attrition rate of ~16% is moderate but costly at scale — losing roughly 1 in 6 employees per cycle drives continual recruiting, onboarding, and productivity-ramp costs, especially since attrition is concentrated among newer and mid-career staff (see Section 6).

4. Attrition by Department & Job Role

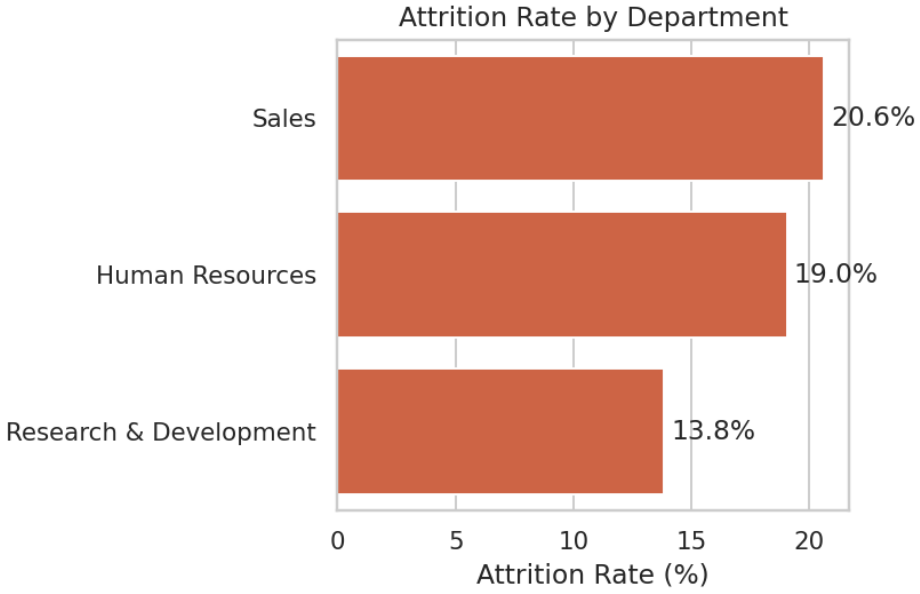


Figure 2: Sales (20.6%) and HR (19.1%) have the highest departmental attrition; R&D is lowest (13.8%). However, because R&D is by far the largest department, it still accounts for 56% of all leavers in absolute terms.

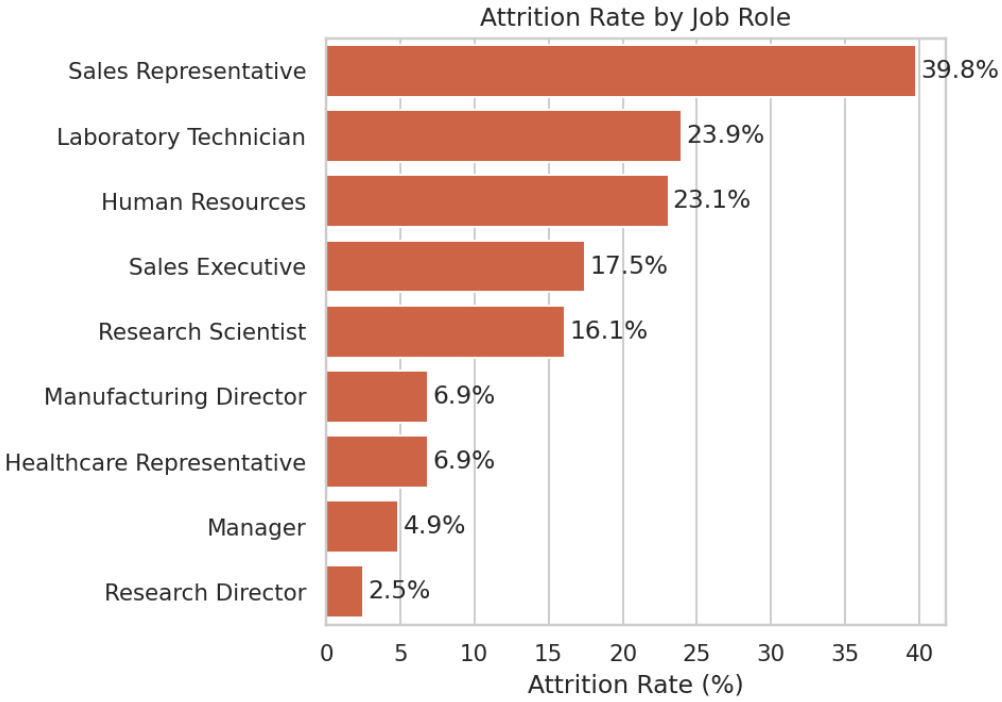


Figure 3: Sales Representatives have by far the highest attrition rate (39.8%), followed by Laboratory Technicians (23.9%) and HR staff (23.1%). Managers, Research Directors, and Manufacturing Directors are the most stable roles (<7%).

Insight: Attrition risk is not evenly spread by department size — it is concentrated in specific, junior, high-churn roles (Sales Representative, Lab Technician). These two roles alone account for a disproportionate share of total attrition relative to their headcount.

5. OverTime, Business Travel & Work-Life Balance

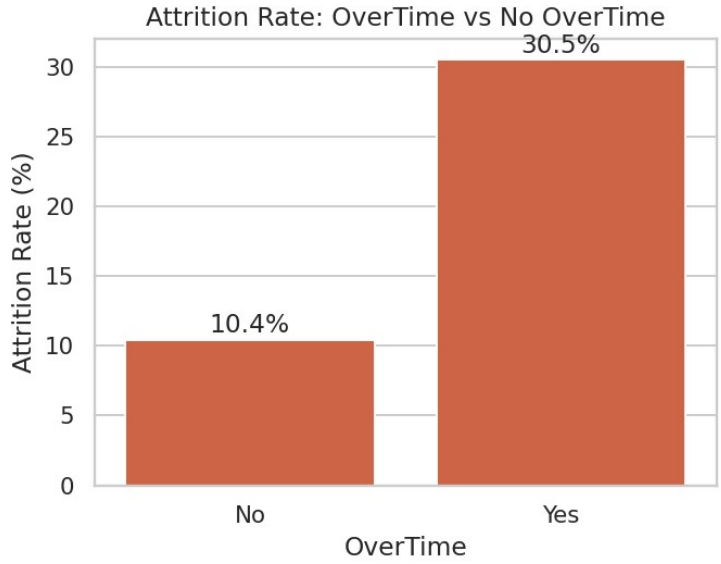


Figure 4: Employees who work overtime leave nearly 3x more often (30.5%) than those who don't (10.4%) — the single strongest categorical driver of attrition in the dataset.

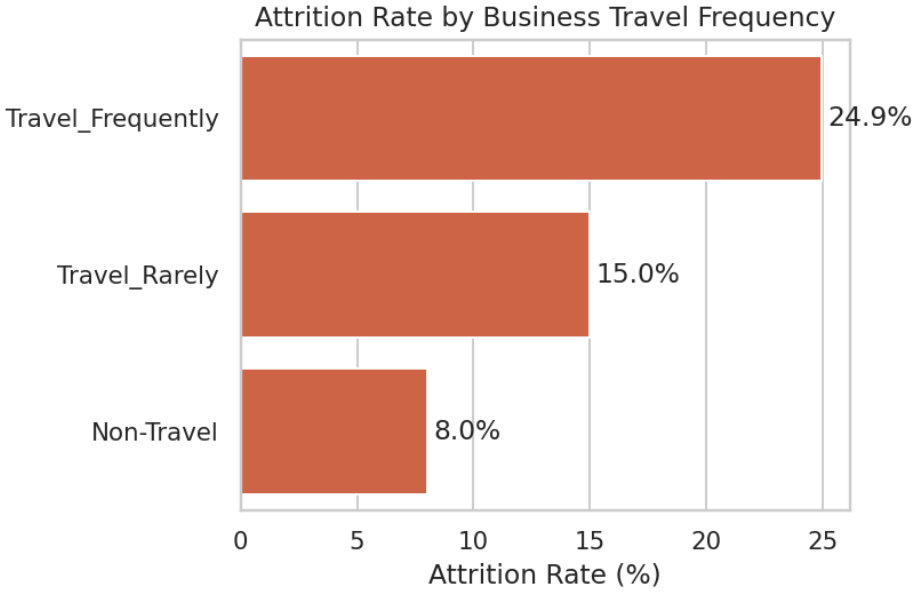


Figure 5: Frequent travelers churn at 24.9% vs 8.0% for non-travelers.

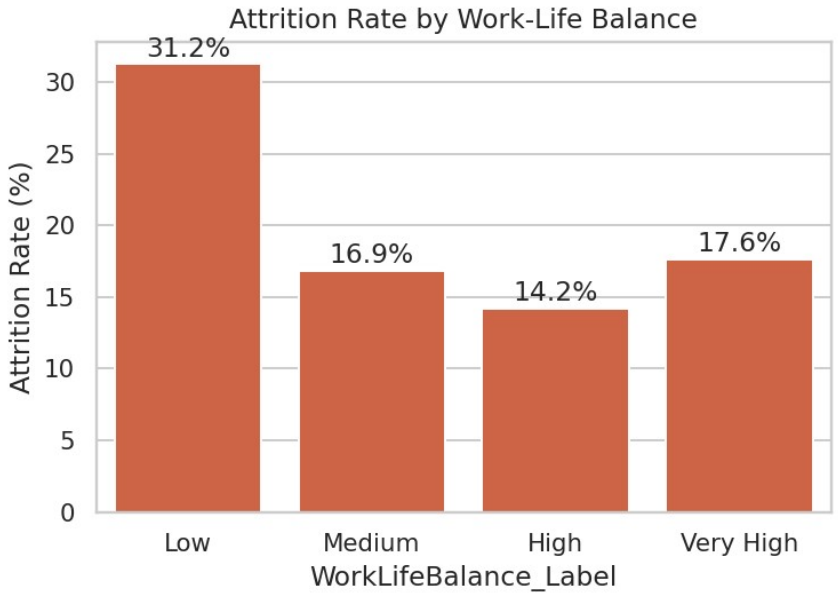


Figure 6: Employees rating their work-life balance "Low" leave at 31.3%, more than double the rate of those rating it "High" (14.2%).

Compounding effect: Single employees who also work overtime leave at 49.6% — roughly 1 in 2 — versus the 16.1% company average. Sales Representatives who work overtime leave at 66.7%. These overlapping risk factors point to workload and role/lifestyle mismatch as the dominant attrition mechanism, far more than pay alone.

6. Tenure, Age & Income Patterns

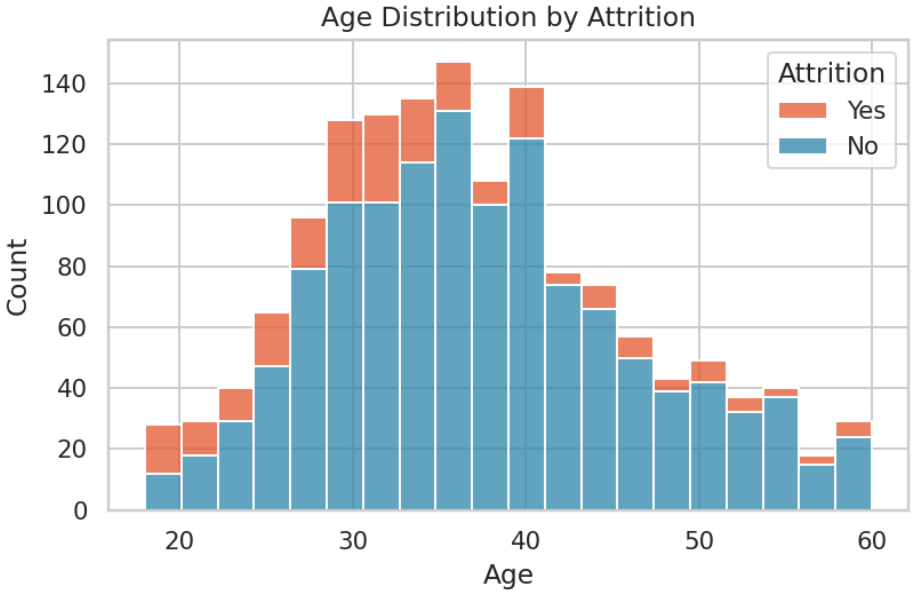


Figure 7: Attrition is heavily weighted toward younger employees; the average age of leavers is 33.6 vs 37.6 for those who stay.



Figure 8: Employees who leave have a median tenure of roughly 2-3 years, well below the ~5-year median for those who stay. Employees with under 2 years' tenure leave at 34.9%, more than double the company average.

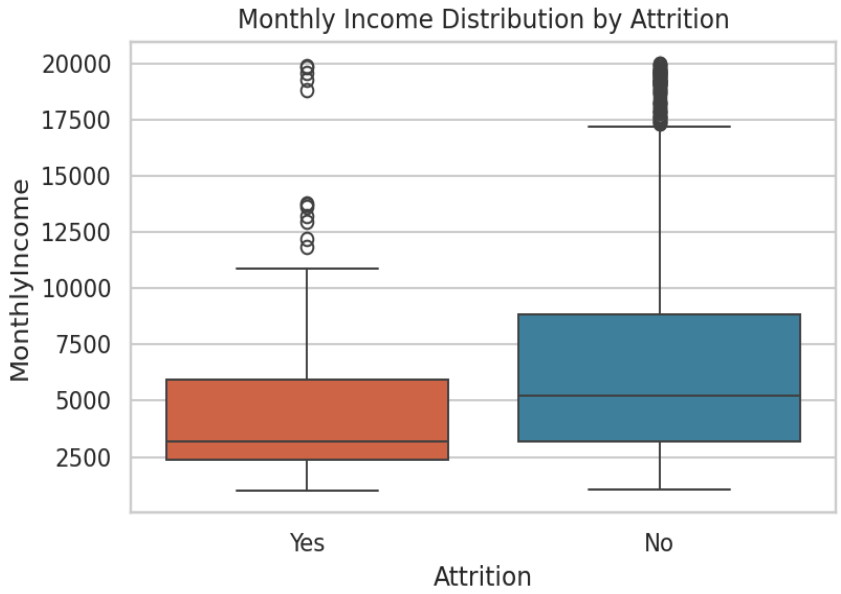


Figure 9: Leavers earn a noticeably lower median monthly income than those who stay (company-wide averages: ₹4,787 vs ₹6,833), consistent with attrition being concentrated at Job Level 1 (26.3% attrition) and entry-level roles.

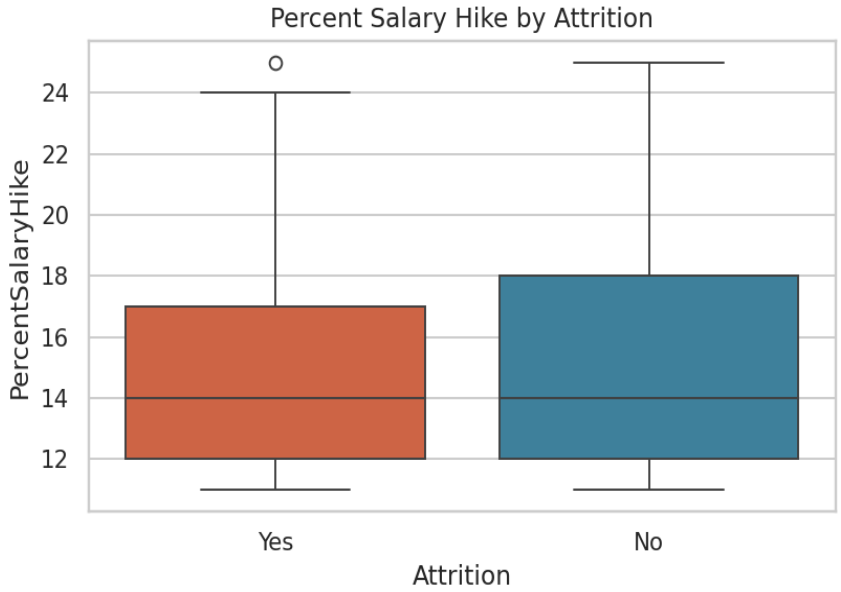


Figure 10: Interestingly, percent salary hike shows almost no difference between leavers and stayers (~15.1% vs ~15.2%) — the size of the last raise is not a meaningful predictor on its own.

7. Satisfaction, Involvement & Stock Options

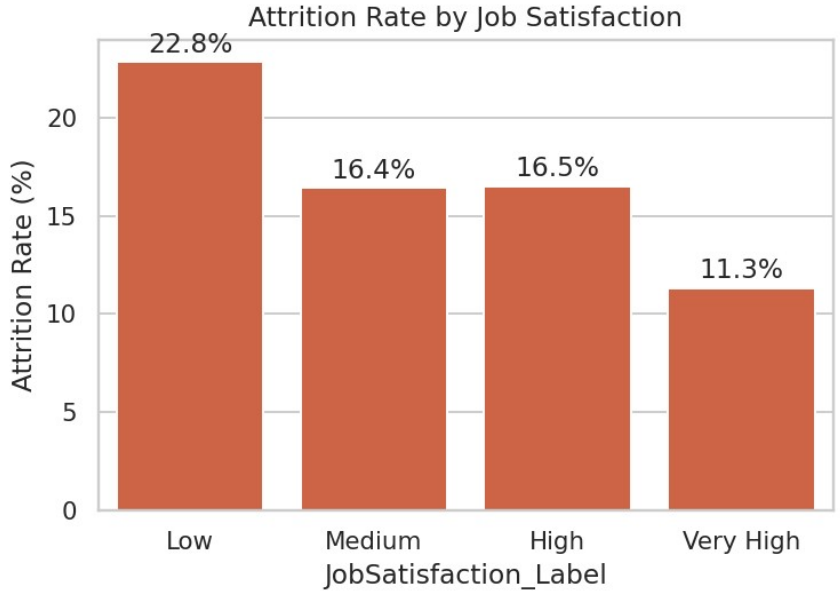


Figure 11: Job satisfaction shows a clear gradient — "Low" satisfaction employees leave at 22.8% vs 11.3% for "Very High".

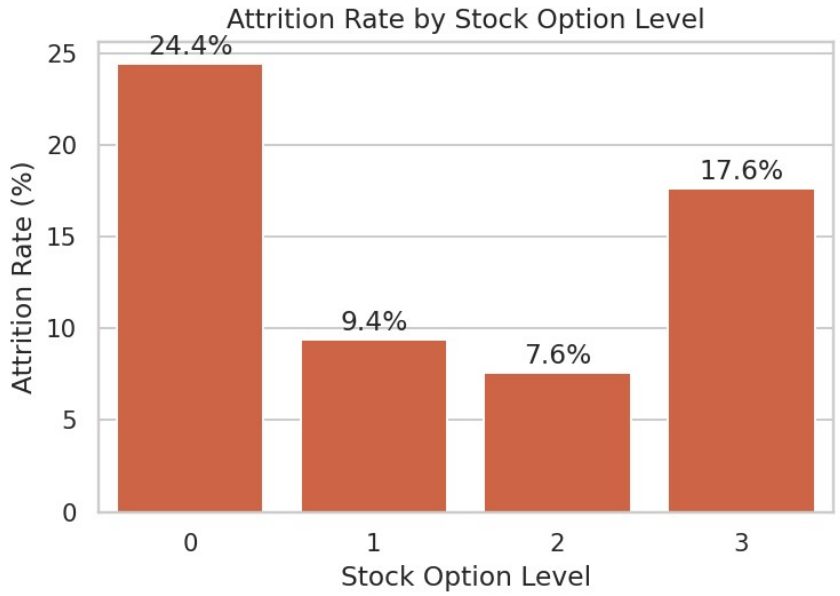


Figure 12: Employees with zero stock options leave at 24.4%, roughly 3x the rate of employees with Level 1 or 2 stock options (~8-9%), suggesting equity-based incentives are strongly associated with retention (levels 1-2 specifically; level 3 is a small, mixed group).

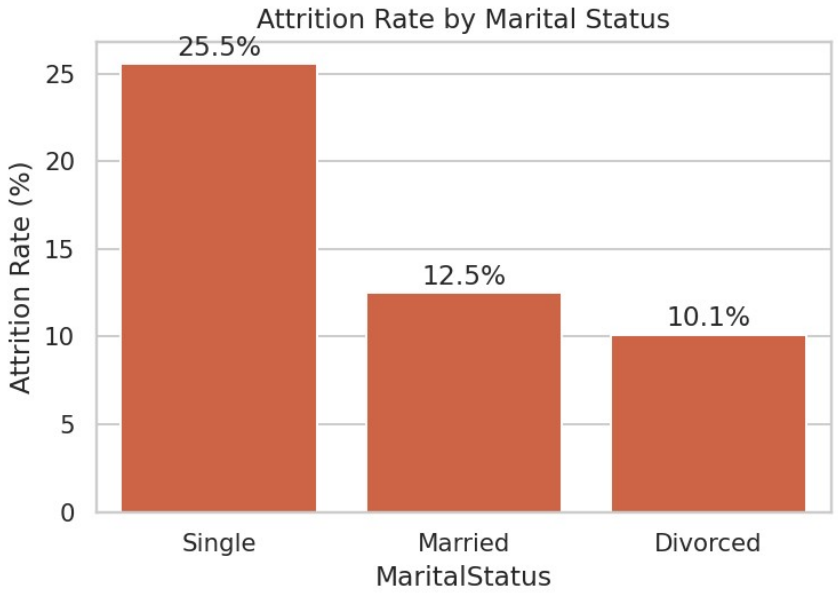


Figure 13: Single employees leave at 25.5%, versus 12.5% for married and 10.1% for divorced employees — consistent with the overtime/lifestyle-flexibility finding above.

8. Correlation Analysis

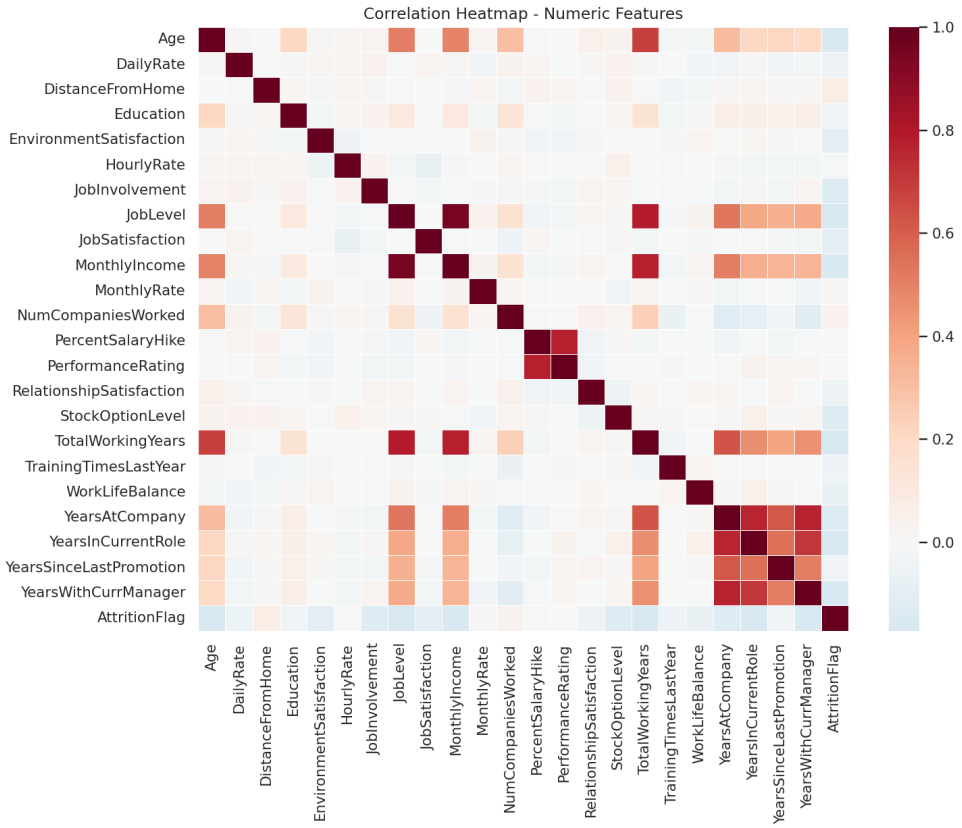


Figure 14: Correlation heatmap across numeric/ordinal features.

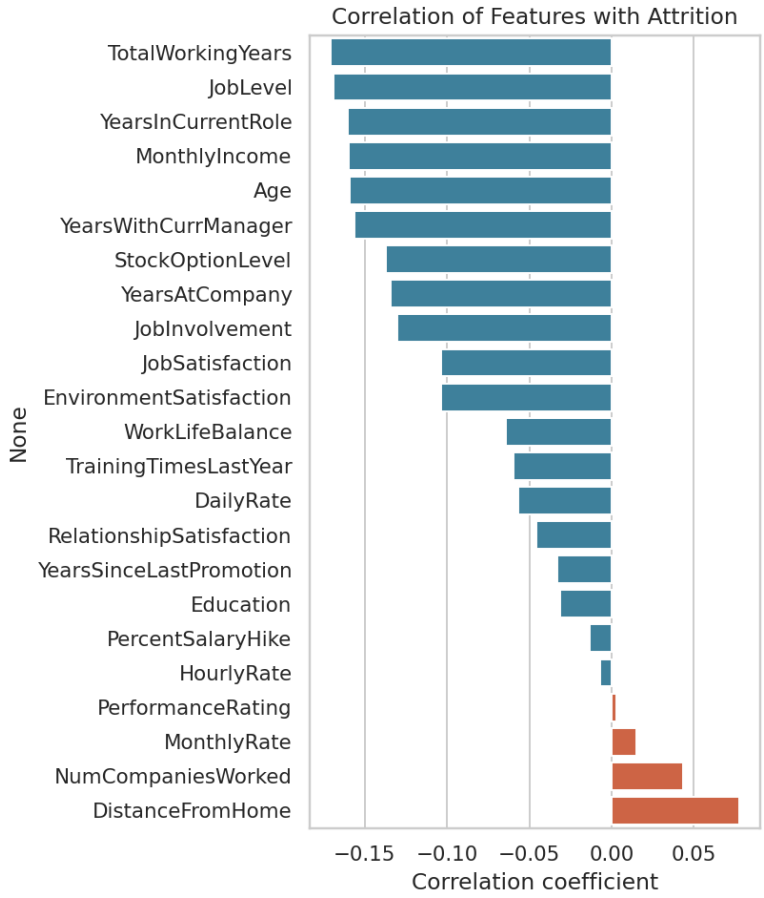


Figure 15: Features most negatively correlated with attrition (i.e., higher values → lower attrition) are TotalWorkingYears, JobLevel, YearsInCurrentRole, MonthlyIncome, and Age. DistanceFromHome is the only feature with a meaningfully positive correlation with attrition, though all correlations here are modest (|r| < 0.2), confirming that attrition results from a combination of factors rather than any single one.

9. Key HR Metrics (KPI Summary)

KPI	Value
Overall Attrition Rate	16.12%
Retention Rate	83.88%
Attrition Rate — OverTime employees	30.53%
Attrition Rate — Non-OverTime employees	10.44%
Attrition Rate — Single employees	25.53%
Attrition Rate — Employees with <2 yrs tenure	34.9%
Attrition Rate — Job Level 1 (entry level)	26.34%
Attrition Rate — Zero Stock Options	24.41%
Attrition Rate — High Performers (Rating = Outstanding)	16.4% (in line with company average — high performers are not disproportionately retained)
Avg. Training Sessions/Year — Leavers vs Stayers	2.6 vs 2.8 (leavers received slightly less training)
Avg. Companies Worked Previously — Leavers vs Stayers	2.9 vs 2.6 (leavers have a slightly more mobile work history)

10. Summary of Key Insights

- **Workload is the single biggest driver: overtime workers leave at nearly 3x the rate of others, and this compounds sharply for single employees (49.6%) and Sales Representatives (66.7%) working overtime.**

- Attrition is a "new employee" problem as much as anything else: over a third of employees with under two years of tenure leave, and average tenure of leavers is barely half that of stayers.
- Entry-level and specific roles carry outsized risk: Sales Representatives (39.8%), Lab Technicians (23.9%), and Job Level 1 staff (26.3%) drive a disproportionate share of total attrition.
- Compensation structure matters more than the size of raises: lower absolute income and zero stock options are strongly linked to attrition, but the percentage salary hike itself is not — meaning one-off raises are unlikely to fix this on their own.
- Satisfaction and involvement scores (job satisfaction, environment satisfaction, work-life balance, job involvement) all show consistent, monotonic relationships with attrition — the lower the score, the higher the churn — making these useful early-warning indicators.
- Frequent travel roughly triples attrition risk (24.9% vs 8.0% for non-travelers), suggesting travel demands are a meaningful, under-addressed strain.
- **High performers are not being retained any better than average performers (16.4% vs 16.1%), which is a warning sign — the company’s best people are leaving at essentially the same rate as everyone else.**

11. Business Recommendations

11.1 Immediate / Low-Cost Actions

- Audit and cap chronic overtime, especially in Sales and Laboratory roles — this is the single highest-leverage lever identified in the data.
- Build a structured 0-24 month onboarding/retention program (mentorship, check-ins at 30/60/90/180 days) targeting the 34.9% attrition rate among new hires.
- Introduce or expand a baseline stock-option grant at Job Level 1-2, where zero-option employees show ~3x the attrition of option-holders.
- Flag employees who score "Low" on Job Satisfaction, Environment Satisfaction, or Work-Life Balance for proactive manager check-ins — these scores are reliable early-warning signals.

11.2 Medium-Term Actions

- Redesign the Sales Representative role (workload distribution, travel expectations, or compensation structure) given its 39.8% attrition rate and the extreme 66.7% rate among those who also work overtime.
- Review travel policies for "Travel_Frequently" employees — consider rotation, travel caps, or additional compensation/time-off-in-lieu.
- Build a targeted retention track for high performers (recognition, accelerated growth paths, retention bonuses) since they currently leave at the same rate as average performers.
- Increase training investment slightly across the board — leavers averaged fewer training sessions, and training is linked to engagement and internal mobility.

11.3 Strategic / Longer-Term Actions

- Introduce flexible-work and lifestyle-support policies aimed at single/early-career employees, who show a much higher propensity to leave, likely due to fewer ties keeping them at one employer.
- Build a live attrition-risk dashboard (Section 12) that HR business partners can use to monitor at-risk segments monthly rather than reactively after resignations.
- Establish exit-interview data collection specifically tagged to these risk factors (overtime, tenure, satisfaction) to validate and refine this model over time.

12. Interactive Dashboard

An interactive HR Attrition Dashboard (delivered as a separate HTML file, Attrition_Dashboard.html) accompanies this report. It lets stakeholders filter by Department, Job Role, OverTime status, and Gender, and view live-updating KPIs and charts covering everything summarized above — functioning as a lightweight, shareable equivalent of a Tableau/Power BI workbook that opens directly in any browser with no software installation required.